

Day 23 Student Handout

Geometry Summer Credit | Modeling With Geometry

Name:	Date:	Period:
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Learning Targets

I can choose assumptions and formulas for a geometry modeling problem. I can use area, volume, scale, and density to make decisions. I can explain whether a model is reasonable.

Lesson 23A: Build a Geometry Model

Warm-Up

1.	A rectangular room is 18 ft by 12 ft. Find the floor area.
2.	A box is 10 in by 8 in by 6 in. Find its volume.
3.	If a model uses scale factor $\frac{1}{4}$, what happens to area? What happens to volume?

Modeling Cycle

1. Understand the situation. 2. Make assumptions. 3. Choose formulas. 4. Calculate. 5. Check units and reasonableness. 6. Revise if needed.

Problem	Work / notes
A classroom wall is 24 ft wide and 9 ft tall. One gallon of paint covers about 350 square feet. How many gallons are needed for one coat?	Assumption: Area: Gallons: Decision:
A planter box is 36 in by 18 in by 12 in. Soil is sold in bags of 1.5 cubic feet. How many bags are needed?	Convert or compare units: Volume: Bags: Decision:
A scale model of a storage tank has volume 20 cubic inches. The actual tank has scale factor 6 from the model. Find actual volume.	Volume scale: Actual volume:

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Lesson 23B | Design Task and Reasonableness

Desmos-Supported Modeling Task

Design Scenario

You are designing a cylindrical container. You may use Desmos to test radius and height values. Your container must hold at least 500 cubic centimeters while using a reasonable amount of material.

Problem	Work / notes
Design A has radius 4 cm and height 10 cm. Find its volume and decide if it meets the 500 cm ³ capacity constraint.	Volume: Compare: Decision:
Design B has radius 5 cm and height 7 cm. Estimate its surface area in terms of pi and compare material use.	Surface area formula: Substitute: Interpret:
Name one assumption that makes the container model simpler but less realistic.	Assumption: Why it helps: What it ignores:

4.	Choose a radius and height for a cylinder that holds at least 500 cm ³ . Show your volume check.
5.	Explain one reason your design is reasonable and one limitation of your model.